

Problem 7

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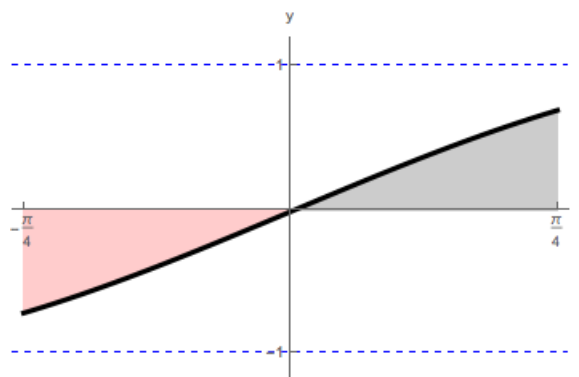
Problem 1 Evaluate the following integrals using symmetry arguments.

(a) $\int_{-\pi/4}^{\pi/4} \sin(t) dt$

Explanation. Since the function $f(t) = \sin(t)$ is odd, $\int_{-\pi/4}^{\pi/4} \sin(t) dt = 0$

We can illustrate our computation with the graph of the function f , where

$$f(t) = \sin t, \text{ for } -\frac{\pi}{4} \leq t \leq \frac{\pi}{4}.$$



(b) $\int_{-2}^2 (1 + x + 3x^7 - x^9) dx$

Explanation. In order to take advantage of symmetry, we will split the given polynomial into a sum of its even and odd parts. We will then integrate each part separately:

$$\int_{-2}^2 (1 + x + 3x^7 - x^9) dx = \int_{-2}^2 1 dx + \int_{-2}^2 (x + 3x^7 - x^9) dx$$

We will first integrate an even function:

$$\int_{-2}^2 1 dx = 2 \int_0^2 1 dx = 2(2) = 4$$

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Then an odd function:

$$\int_{-2}^2 (x + 3x^7 - x^9) dx = 0$$

$$\text{Therefore: } \int_{-2}^2 (1 + x + 3x^2 - x^9) dx = 4$$

(c) $\int_{-\pi}^{\pi} x \cos(x) dx$

Explanation. $\int_{-\pi}^{\pi} x \cos(x) dx = 0$, since the product of an odd and even function is odd.

We can illustrate our computation with the graph of the function f , where

$$f(x) = x \cos x, \text{ for } -\pi \leq x \leq \pi.$$

